Power



File: power.kicad_sch

Vehicle (OBD, CAN, K-Line)



File: vehicle.kicad_sch

MCU (ESP32-C5, USB)



File: mcu.kicad_sch

LTE + GNSS (EC200U)



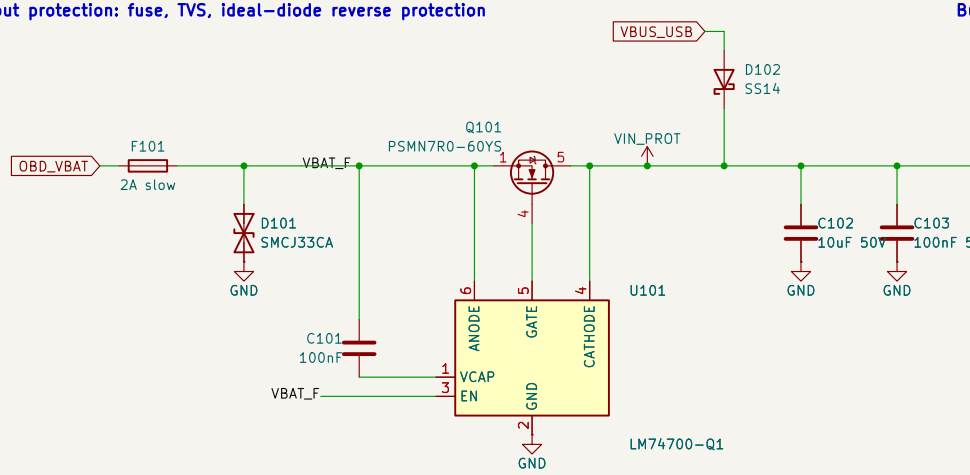
File: lte.kicad_sch

Sheet: /
File: obd-v2x-dongle.kicad_sch

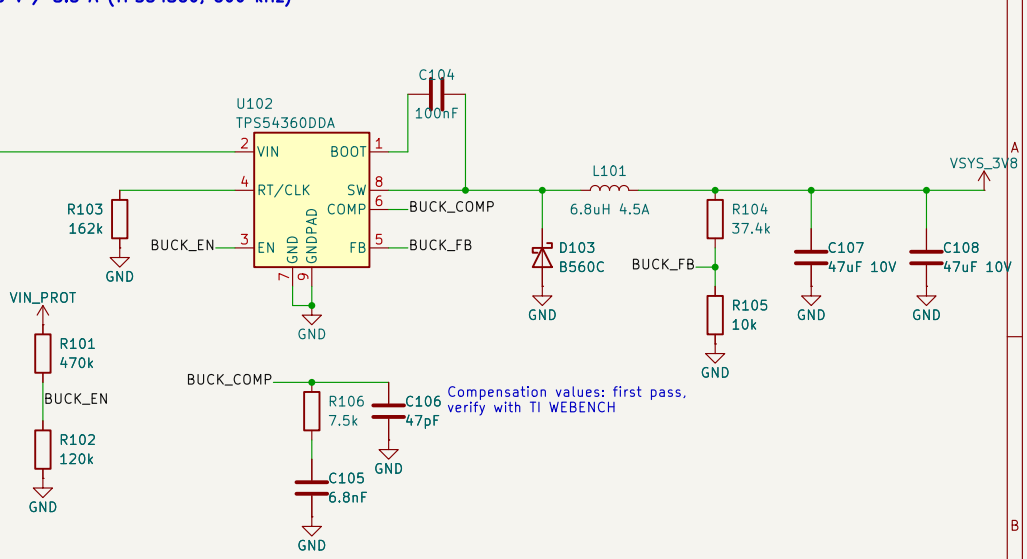
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/5

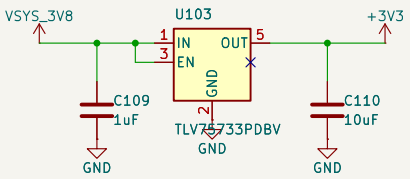
Input protection: fuse, TVS, ideal-diode reverse protection



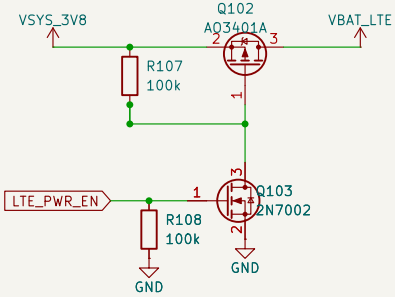
Buck 3.8 V / 3.5 A (TPS54360, 600 kHz)



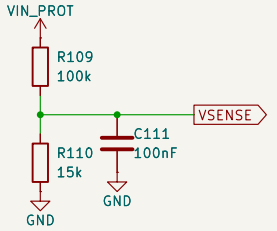
LDO 3.3 V (logic)



LTE modem supply switch (LTE_PWR_EN high = on)



Battery sense (VSENSE = VIN x 0.13)



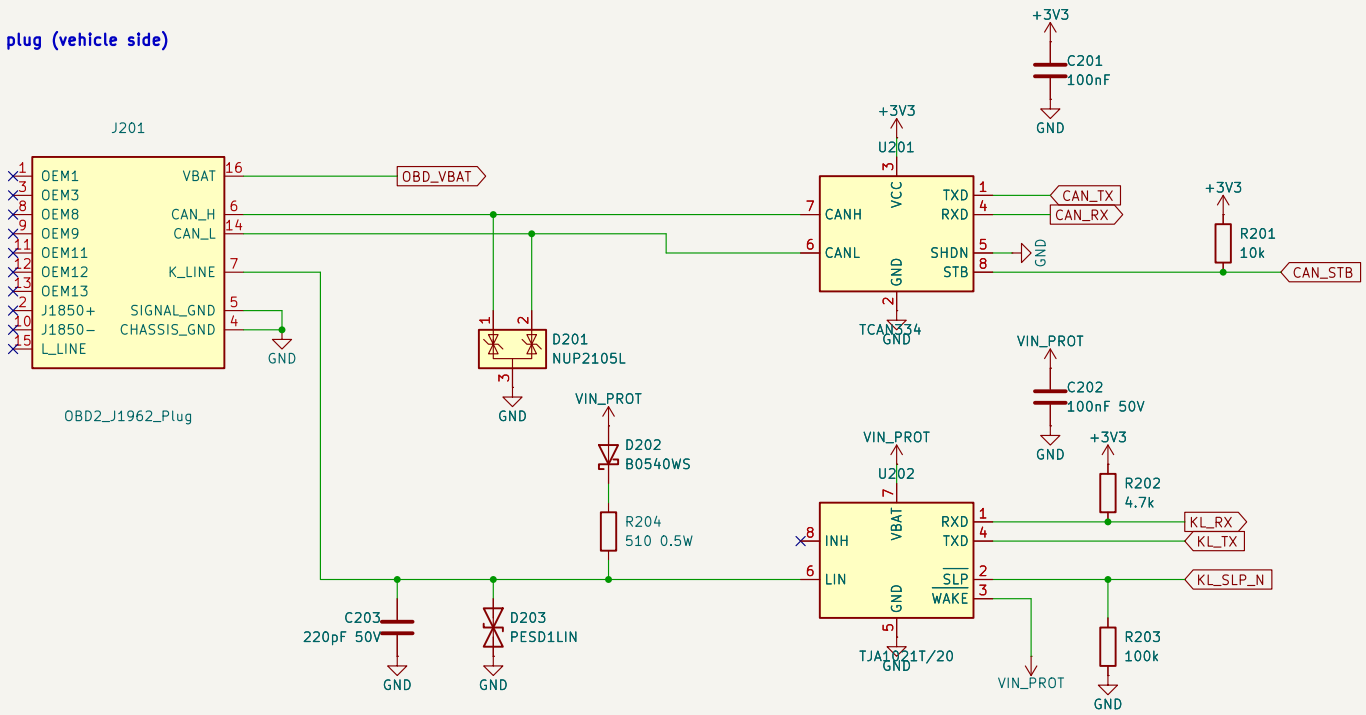
ERC power flags



Sheet: /Power/	
File: power.kicad_sch	
Title: Power: protection, buck 3.8 V, LDO 3.3 V	
Size: A4	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.1**
	Id: 2/5

High-speed CAN (ISO 15765), no termination in the dongle

OBD-II plug (vehicle side)



K-Line (ISO 9141 / KWP2000) via LIN transceiver + 510R tester pull-up

Sheet: /Vehicle (OBD, CAN, K-Line)/	
File: vehicle.kicad_sch	
Title: Vehicle interface: OBD-II, CAN, K-Line	
Size: A4	Date:
KiCad E.D.A. 10.0.8	Rev: 0.1
	Id: 3/5

ESP32-C5: 5.9 GHz ITS-G5 receive via U.FL on the module

+3V3
C302
10uF
GND
+3V3
C303
100nF
GND

+3V3
R301
10k
SW301
RESET
GND
C301
1uF
GND

+3V3
U301
ESP32-C5-WROOM-1U-N16R8

2	3V3	IO11/TXD0	25
3	EN	IO12/RXD0	24
		IO13/USB_D-	13
		IO14/USB_D+	14
6	IO0/XTAL_32K_P	IO23	21
7	IO1/ADC1_CH0	IO24	23
4	IO2/MTMS	IO25	26
5	IO3/MTDI	IO26	27
17	IO4/MTCK	IO27	18
16	IO5/MTDO	IO28	15
8	IO6		
9	IO7	ANT	31
10	IO8	NC	20
11	IO9	NC	22
12	IO10		
19	IO15		
	GND		
1	GND		

KL_RX
VSENSE
LTE_PWRKEY_CTL
LTE_RESET_CTL
CAN_TX
CAN_RX
KL_TX
KL_SLP_N
LTE_PWR_EN
LTE_STATUS_3V3
CAN_STB

UART0_TX
UART0_RX
USB_DN
USB_DP
MCU_TX_LTE
MCU_RX_LTE
MCU_CTS_LTE
MCU_RTS_LTE
LED_DIN
ESP_BOOT

Boot / status LED

+3V3
R302
10k
ESP_BOOT
SW302
BOOT
GND

VSYS_3V8
C304
100nF
GND

VSYS_3V8
LED_DIN
DIN
VDD
DOUT
VSS
D301
SK6812MINI
GND

USB-C: flashing / bench power

J301
USB-C
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
SH
A4
A5
B5
A7
B7
A6
B6
A8
B8
A1
VBUS_USB
R303
5.1k
R304
5.1k
GND
USB_DN
USB_DP
GND

U302
USB_DP
USB_DN
USB_DP
USB_DN
GND
USBLC6-2SC6

UART0 test pads

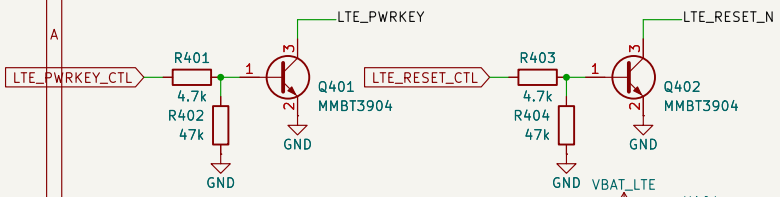
UART0_TX
TP301
UART0_TX
UART0_RX
TP302
UART0_RX
GND
TP303
GND

Sheet: /MCU (ESP32-C5, USB)/
File: mcu.kicad_sch

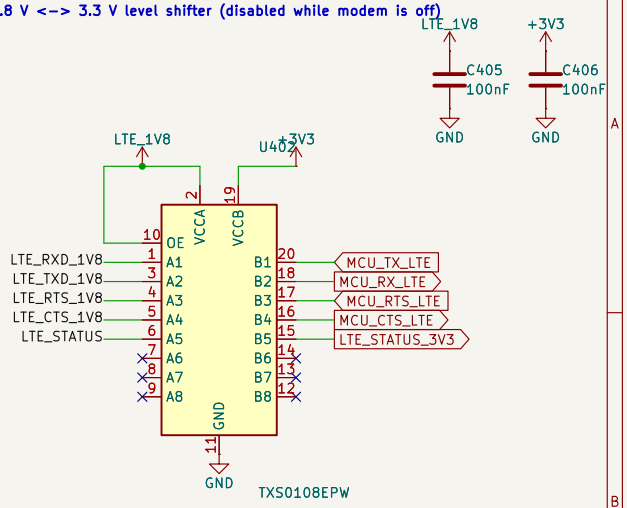
Title: MCU: ESP32-C5 (ITS-G5 RX, CAN FD, BLE), USB-C

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 4/5

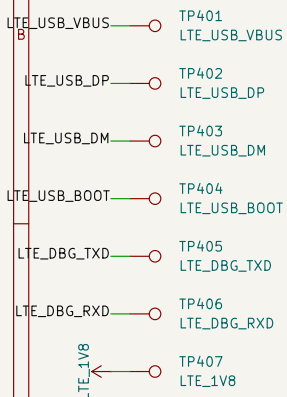
PWRKEY / RESET_N open-collector drivers (MCU high = pull low)

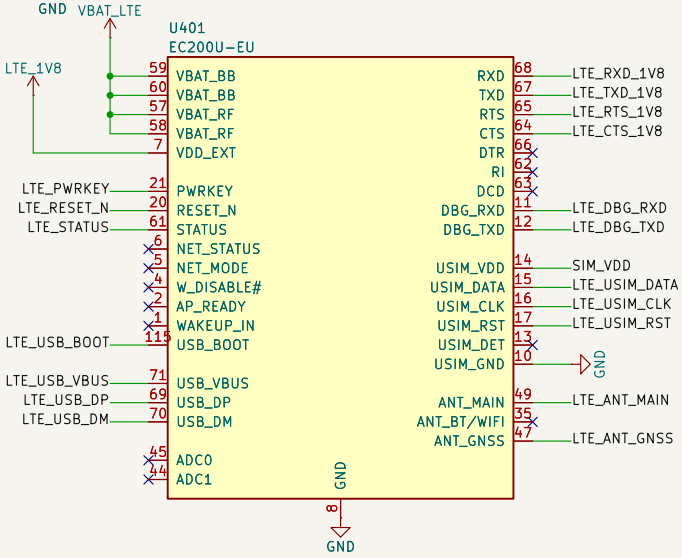


1.8 V <-> 3.3 V level shifter (disabled while modem is off)

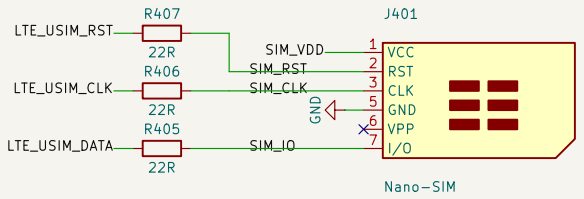


Test pads: modem USB (firmware update), debug UART

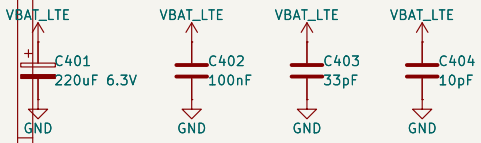


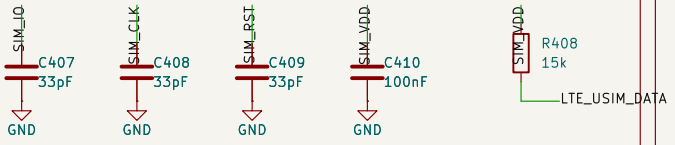


Nano-SIM

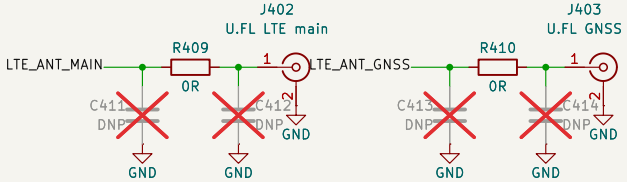


VBAT_LTE bulk/decoupling (close to pins 57-60)





Antennas (pi-match, shunt caps DNP until tuned)



Sheet: /LTE + GNSS (EC200U)/
File: lte.kicad_sch

Title: LTE Cat 1 + GNSS: Quectel EC200U-EU (1.8 V I/O)

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 5/5